Project Apollo: Modern Threat Intelligence Pipeline

Engineering Architecture & Technical Delivery Report | Q1 Milestone

Executive Summary

Project Apollo delivers an asynchronous, low-latency microservices pipeline designed to ingest, normalize, and score external threat telemetry feeds. This engineering report outlines the architectural choices, benchmarking metrics, and production readiness findings gathered during the Phase II performance testing cycles.

Core Architectural Components

The architecture consists of three containerized subsystems deployed across a high-availability AWS EKS cluster: an Apache Kafka event ingestion bus, a Python FastAPI stream processor with spaCy NLP pipelines, and a TimescaleDB analytical warehouse.

Component	Technology	Replica Count	Target Latency SLA
Ingestion Gateway	Go / Fiber	4 Pods	< 15 ms
NLP Parser	Python / spaCy 3.8	8 Pods	< 120 ms
Storage Tier	PostgreSQL / Timescale	Primary + 2 Read	< 45 ms
Cache / Queue	Redis Enterprise	3-node Cluster	< 5 ms

Methodology and Implementation Details

Data ingestion adheres to the STIX 2.1 JSON standard. Inbound telemetry is validated against JSON schema definitions, sanitized for illegal control characters, and pushed to partitioned Kafka topics organized by source feed.

Stream Processing & Parsing Efficiency

Text tokenization benchmarks achieved 1,450 documents per second per worker utilizing multi-process worker pools. Memory utilization remained stable below 750 MB per container under simulated peak volumetric loads of 10,000 requests/sec.

Database Partitioning Strategy

Telemetry indicators are partitioned by weekly intervals with automated tiering to Amazon S3 Glacier after 90 days. B-tree indices on normalized indicator hashes ensure sub-50ms lookup times for active SOC analyst triage queries.

Verification Results & Production Sign-off

Automated integration tests executed over 500 simulated scenarios with zero unhandled exceptions. Chaos engineering experiments validated automated pod self-healing within 12 seconds of forced node termination.

Testing Metrics Table

Test Scenario	Iterations	Pass Rate	Mean Response
Volumetric Burst (10k req/s)	5 Runs	100.0%	38 ms
Malformed Payload Rejection	1,200 Cases	100.0%	4 ms
Database Failover Simulation	3 Drills	100.0%	8.2 s recovery
End-to-End Ingestion Pipeline	50,000 Msgs	99.98%	112 ms

Conclusion & Rollout Approval

Project Apollo has passed all security, resilience, and performance criteria. Production rollout is scheduled for Tuesday, October 20, 2026, during the standard enterprise maintenance window.